

Chapter 10

Credit Market Imperfections: Credit Frictions, Financial Crises, and Social Security

Chapter 10 Topics

- Credit Market Imperfections and Consumption.
- Asymmetric Information and the Financial Crisis.
- Limited Commitment and the Financial Crisis.

Credit Market Imperfections

- Credit market Imperfections:
 - different interest rate for lenders and borrowers
 - borrowing limits
- There exists many reasons for the existence of Credit Market Imperfections. We focus on 2
 - **asymmetric information**: one party has more information about his characteristic than the party it is trading with
 - **limited commitment**: impossible to commit to pay off his debt in the future
- Both played a key role during the 2008-2009 crisis

Credit Market Imperfections

- r_1 : lending interest rate
- r_2 : borrowing interest rate

$$r_2 > r_1$$

- lenders can lend at a lower interest rate than the one faced by borrowers

Asymmetric Information

Definition

- one party has more information than the other party it is trading with

Example

- For example, a used car salesman knows the quality of a particular car, but the buyer doesn't.

Example

- I know how hard I worked today, but my boss doesn't.

Asymmetric Information

Akerlof (1972) - The Market for Lemons

- “Nature” chooses how good a car will be: good or bad with equal probability
- The buyer offers a price, and then the seller accepts or rejects
- A good car is worth \$35,00 to the buyer, but a bad one only \$28,000
- How much might the buyer offer? Which cars are sold?

Asymmetric Information

- The buyer offers \$31,000 (the average quality)
- Sellers of good cars won't sell: their car is worth \$35,000
- Sellers of bad cars sell - their car is worth \$28,000
- The buyer knows this, and thus offers only \$28,000: No good cars are sold in equilibrium!

Asymmetric Information and Financial Crisis

- Financial Intermediary: Lending carried out through banks
 - deposit rate at banks is r_1
 - loan rate is r_2
- Fraction a of borrowers never defaults, fraction $1 - a$ always defaults
- Asymmetric Information:
 - bank cannot tell the good borrowers from the bad ones
 - All good borrowers identical, borrow L
 - Bad borrowers mimic the good ones

Asymmetric Information and Financial Crisis

- Average profit of the bank:

$$\begin{aligned}\pi &= a \times L(1 + r_2) + (1 - a)0 - L(1 + r_1) \\ &= L[a(1 + r_2) - (1 + r_1)]\end{aligned}$$

- In Equilibrium, profits are equal to 0:

$$1 + r_2 = \frac{1 + r_1}{a}$$

Asymmetric Information and Financial Crisis

$$1 + r_2 = \frac{1 + r_1}{a}$$

- If $a = 1$, $r_2 = r_1$
- Otherwise $r_2 > r_1$

Lifetime Budget Constraint

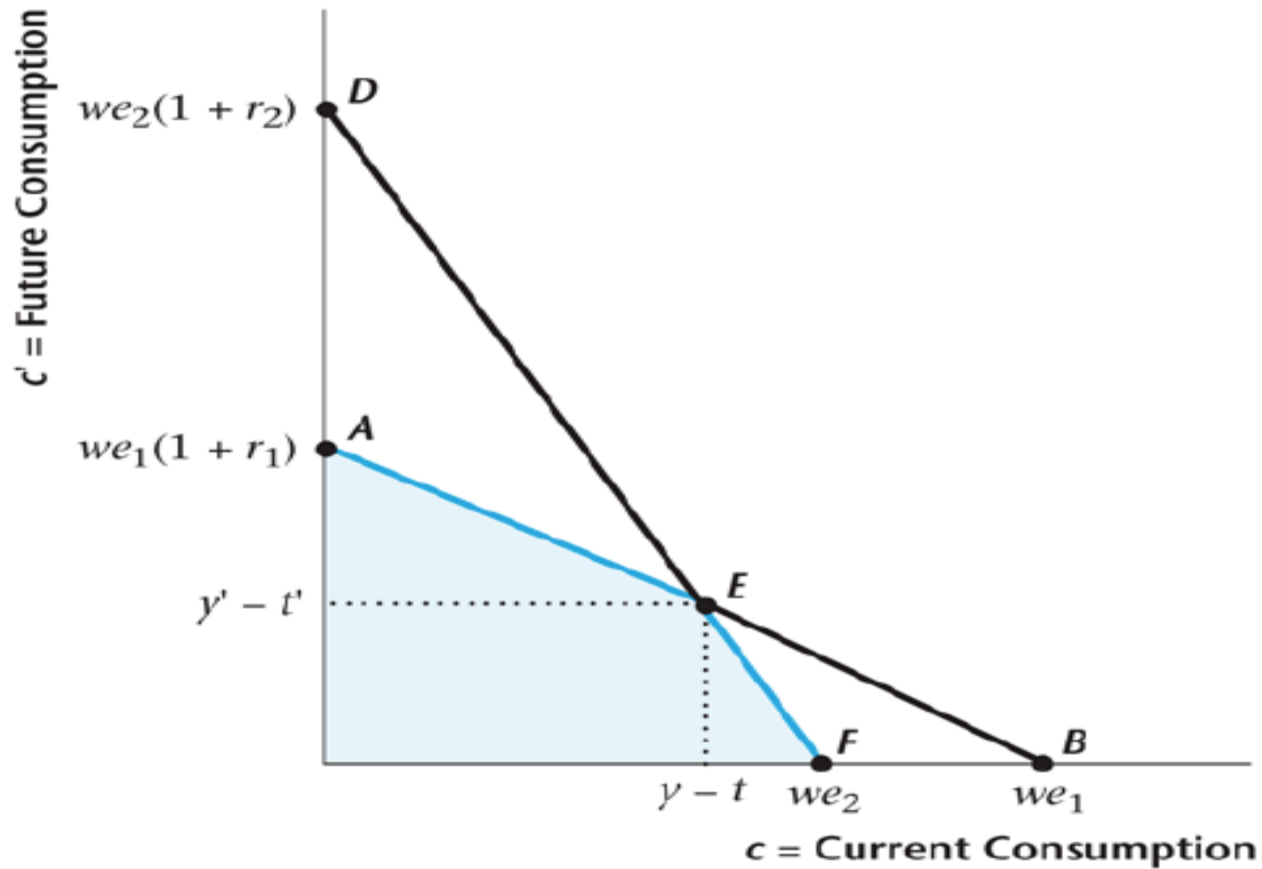
- Lenders:

$$c + \frac{c'}{1 + r_1} = we_1 = y + \frac{y'}{1 + r_1} - t - \frac{t'}{1 + r_1}$$

- Borrowers:

$$c + \frac{c'}{1 + r_2} = we_2 = y + \frac{y'}{1 + r_2} - t - \frac{t'}{1 + r_2}$$

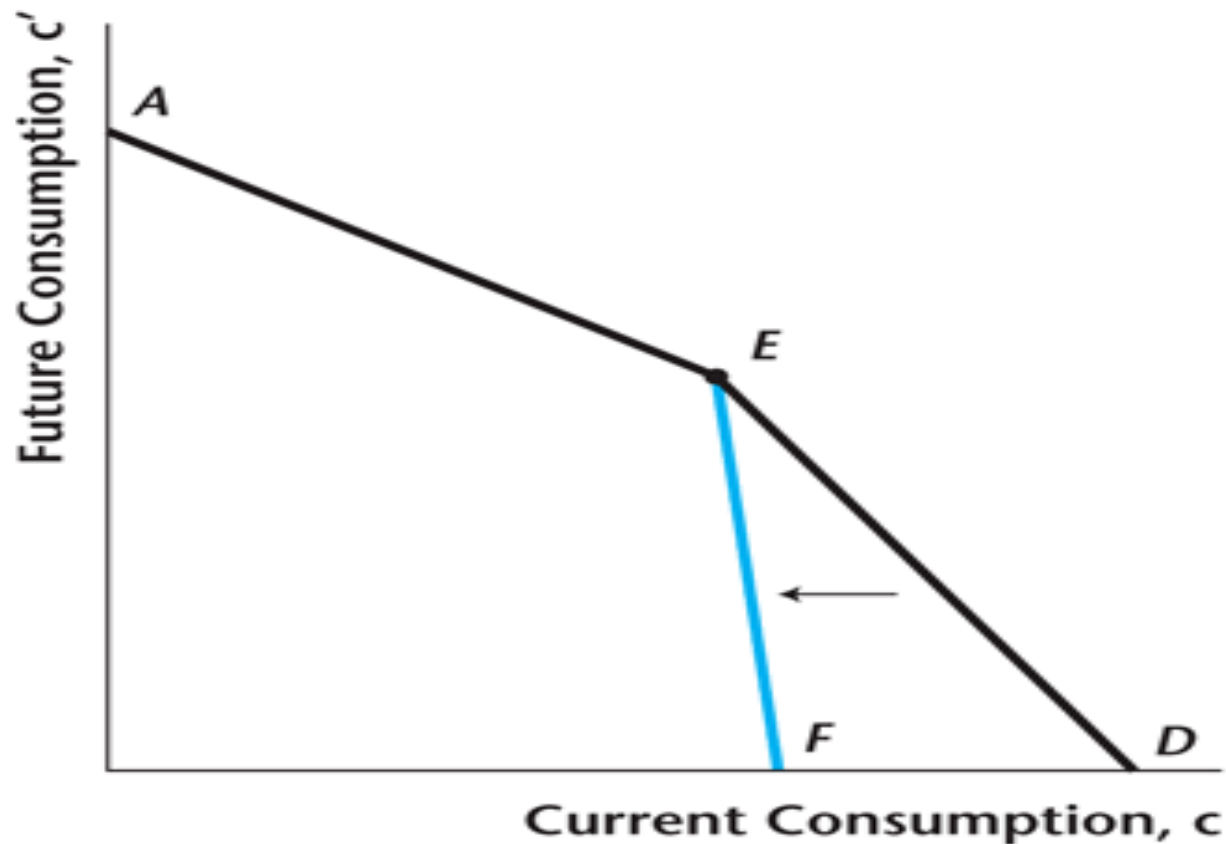
Lifetime Budget Constraint



Effect of a Decrease in the Fraction of Creditworthy Borrowers

- Default premium increases – even good borrowers face higher loan rates.
- Budget constraint shifts in.
- Consumption falls for all borrowers.
 - Substitution Effect: ↓ c : relative price of current consumption increased
 - Wealth Effect: ↓ c Borrower is poorer: higher interest on his debt

Effect of a Decrease in the Fraction of Creditworthy Borrowers



Government

- The government borrows and lends at the interest rate that lenders face
 - treasury securities (US government debt) are backed by full faith and credit of the US Treasury
 - they are considered as risk free
- This implies that Ricardian equivalence does not hold, in general

Effect of a Tax Cut

- Government Budget Constraint:

$$T + \frac{T'}{1 + r_1} = G + \frac{G'}{1 + r_1}$$

- Current Tax cut: $-\Delta t$ with an increasing in future tax: $\Delta t(1 + r_1)$
- It is as if the government was lending to the consumer at the interest rate r_1
- while the consumer would otherwise borrow at $r_2 > r_1$.

Effect of a Tax Cut

- Suppose a consumer initially is credit constrained – that is he or she saves zero.
- For such a consumer, the entire tax cut will be spent on current consumption.
- This is very different from the case with no credit market imperfections, where the consumer will save the entire tax cut to pay higher future taxes.

Effect of a Tax Cut

- Recall: $-\Delta t(1 + r_1) = \Delta t'$

- Impact on lenders:

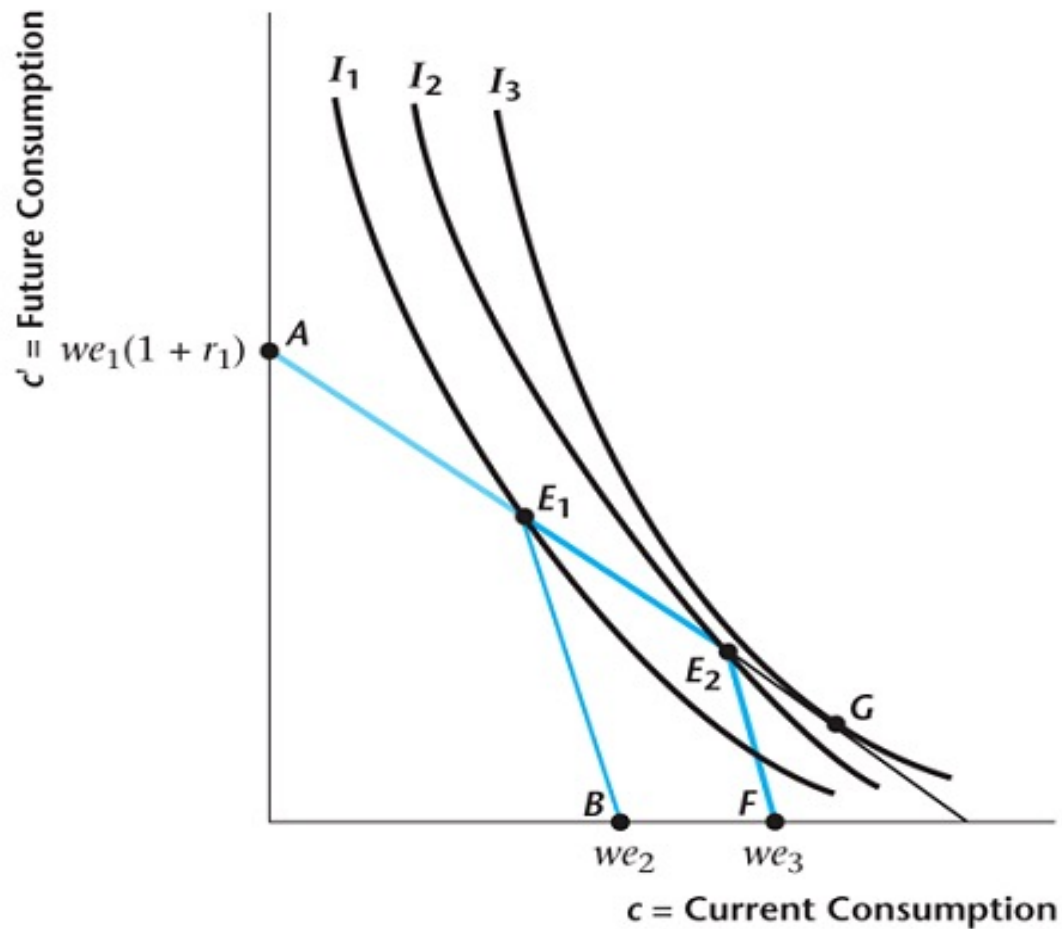
$$\Delta we_1 = -\Delta t - \frac{\Delta t'}{1 + r_1} = -\Delta t + \Delta t = 0$$

- Impact on borrowers:

$$\Delta we_2 = -\Delta t - \frac{\Delta t'}{1 + r_2} = -\Delta t \left(1 - \frac{1 + r_1}{1 + r_2}\right)$$

- Since $r_1 < r_2$, a tax cut $\Delta t < 0$ has a positive wealth effect for borrowers.

Effect of a Tax Cut



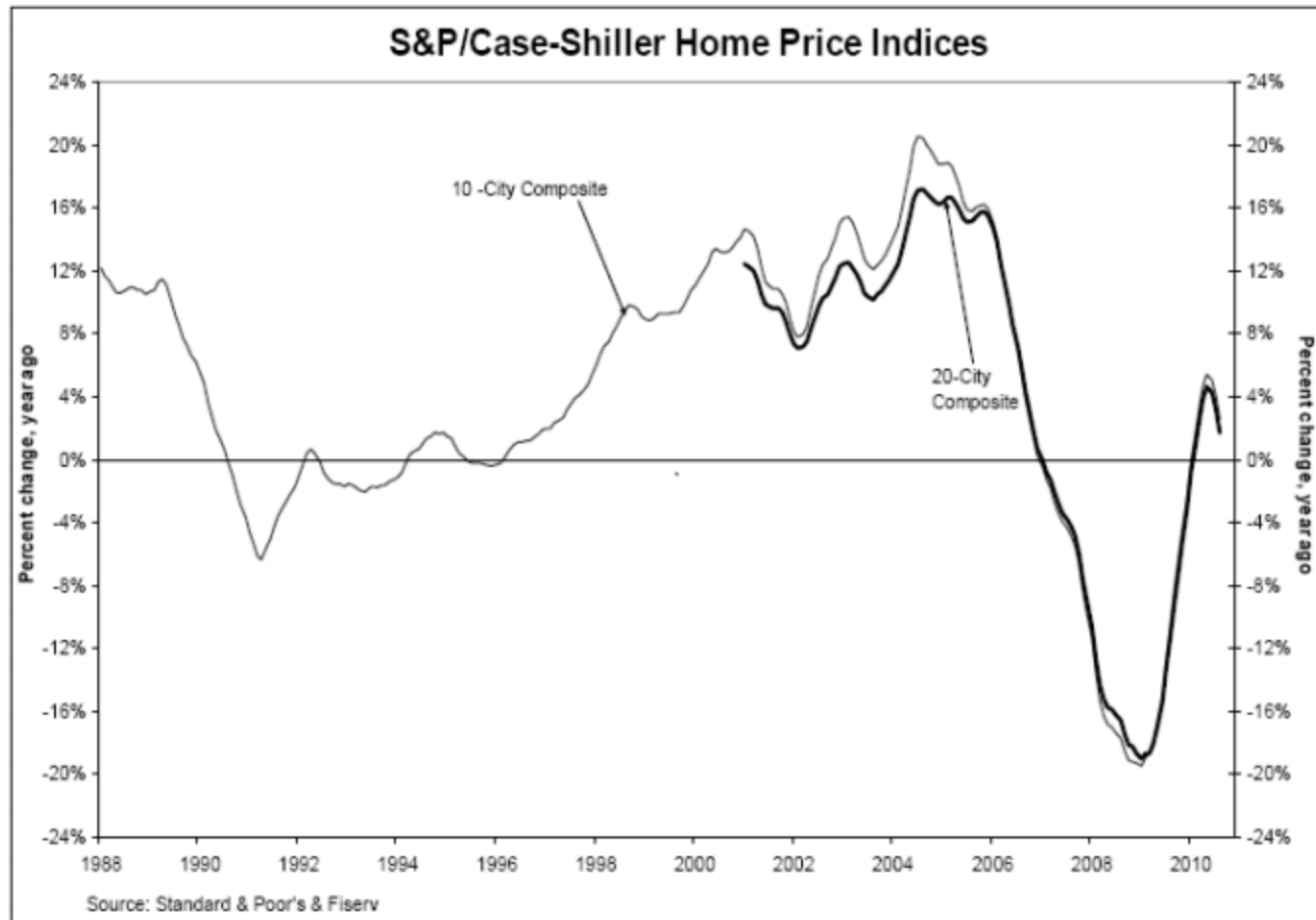
Limited Commitment

- Impossibility for a borrower to commit to pay off his debt
- Borrowers need incentives not to default on their debts
- Incentive device: **collateral** requirement
- Definition
 - A collateral is an asset owned by the borrower that the lender has a right to seize if the borrower defaults
- Example
 - House is collateral for a mortgage loan

Financial crisis 2008 - 2009

- burst of the housing bubble
 - in cities like Phoenix, Las Vegas, Miami: prices have fallen by up to 50% from their peak
- amount of collateral of household declined

Financial crisis 2008 - 2009



Limited Commitment

Falling housing prices hurts consumption through 2 channels:

- 1, amount of collateral of household declined
 - homeowners have less collateral
 - limit homeowners ability to borrow
 - They reduce their consumption of durable goods
- 2, negative Housing wealth effect
 - Lower House prices make people poorer: reduce the amount they have to spend over their lifetimes
 - they distribute some of it in the present because of consumption smoothing

Empirically, 1 is more important than 2

Saving with Assets

- Household owns a house
 - p : price of housing
 - H : quantity of housing owned by the consumer
- Household wants to sell his house to consumer more
- but housing is illiquid and can't be sold in the current period

$$we = y - t + \frac{y' - t' + pH}{1 + r}$$

Collateral Constraint

- Collateral constraint: the discounted value of the house is the borrowing limit:

$$-s(1 + r) \leq pH$$

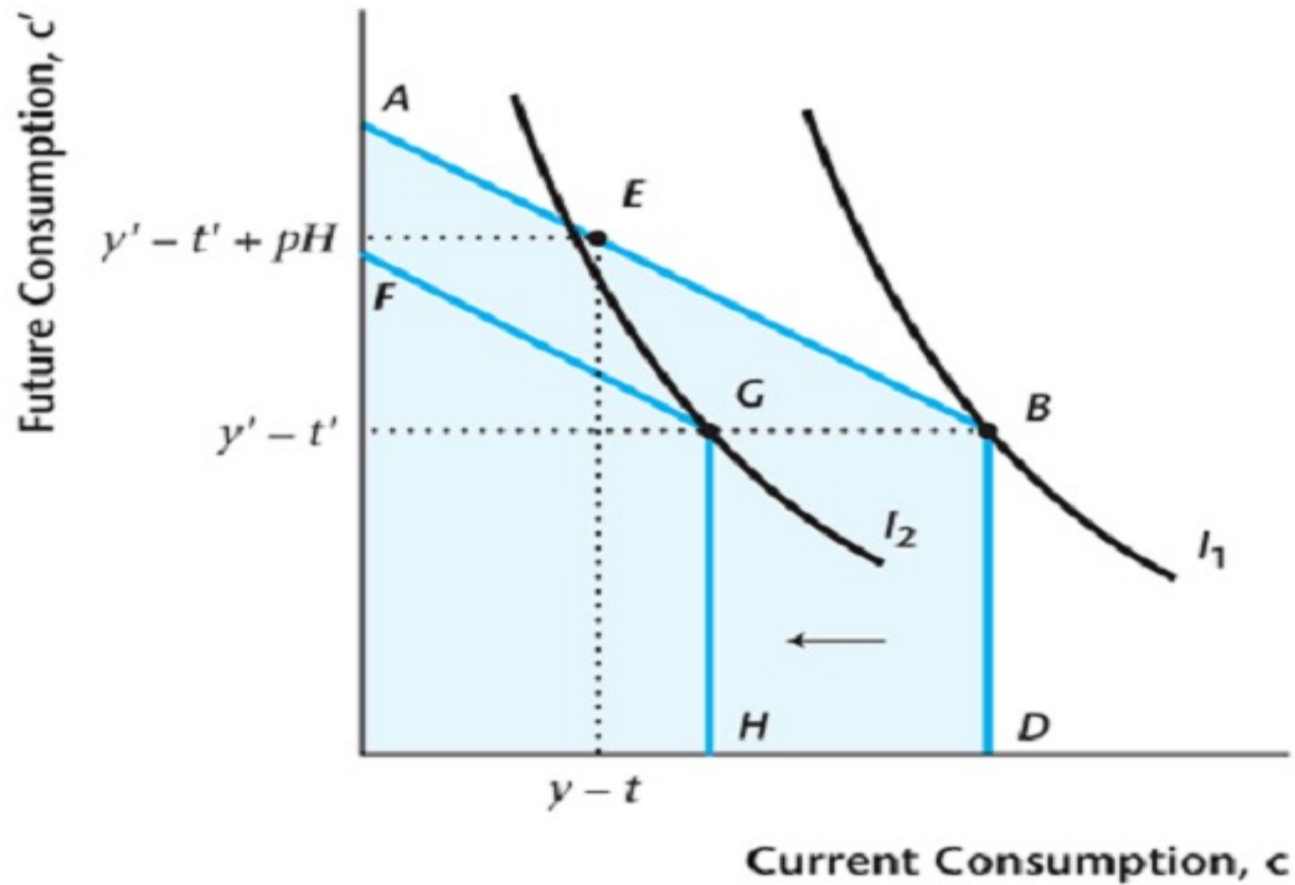
- If the consumer borrows $-s$ today
- he will have to pay off his debt tomorrow with an interest:
 $-s(1 + r)$

Collateral Constraint

- The consumption must satisfy

$$c \leq y - t + \frac{pH}{1+r}$$

Collateral Constraint



Saving with Asset

Financial crisis decreases p :

1. Negative Wealth Effect: budget line shift left
 - c, c' both decrease
2. Collateral Constraint:
 - $-s(1 + r) \leq pH$
 - s increases
 - $\rightarrow c$ decreases and c' increases

This mechanism can explain part of the decrease in durable consumption during the financial crisis.

Pay-as-you-go Social Security

- Taxes on the working population pay for social security transfers to the retired each period.
- Suppose two generations alive at each date, young and old.
- The young pay social security taxes t , the old receive social security benefits b .

Population Growth

- * The population grows according to the following equation. Each period, there are N' young and N old alive.

$$N' = (1 + n)N$$

The Government Balances Its Budget

- * Total social security benefits must equal total taxes on the young.

$$Nb = N't$$

Relationship Between Taxes for the Old and Benefits for the Young

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$$t = \frac{b}{1 + n}$$

Figure 10.8 Pay-As-You-Go Social Security for Consumers Who Are Old in Period T

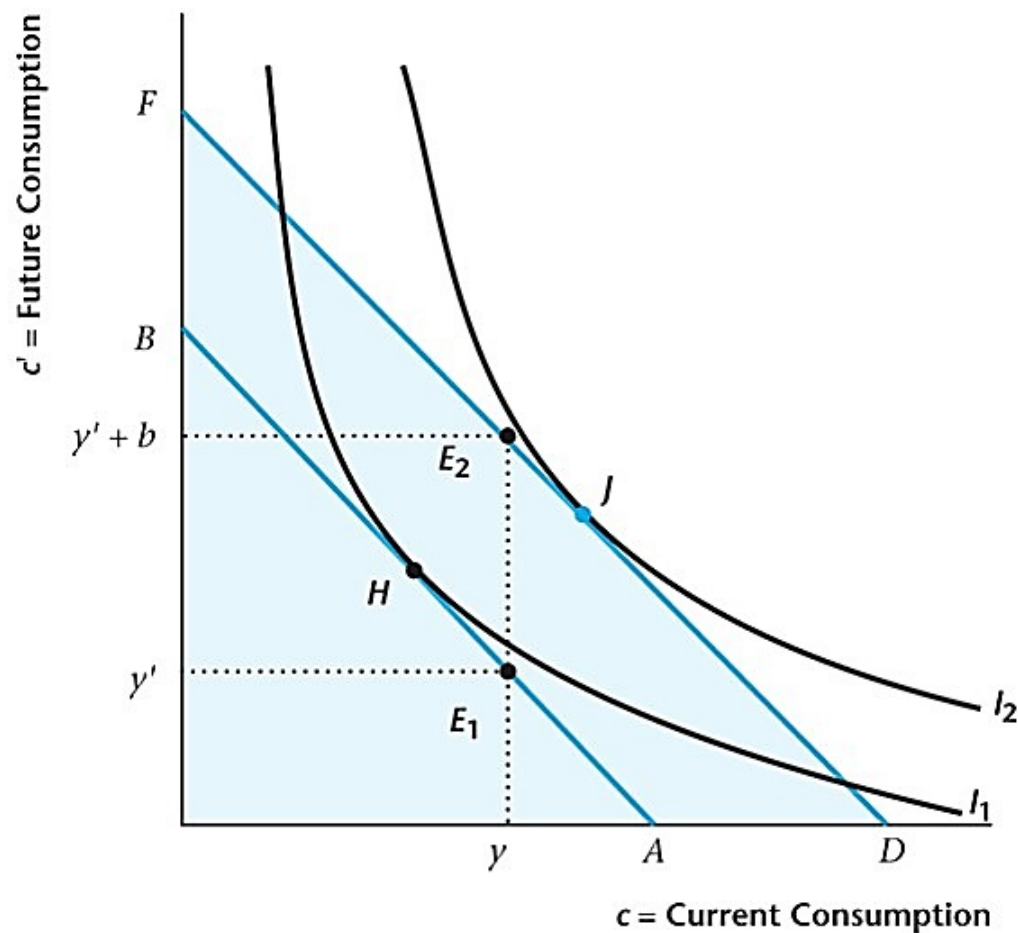
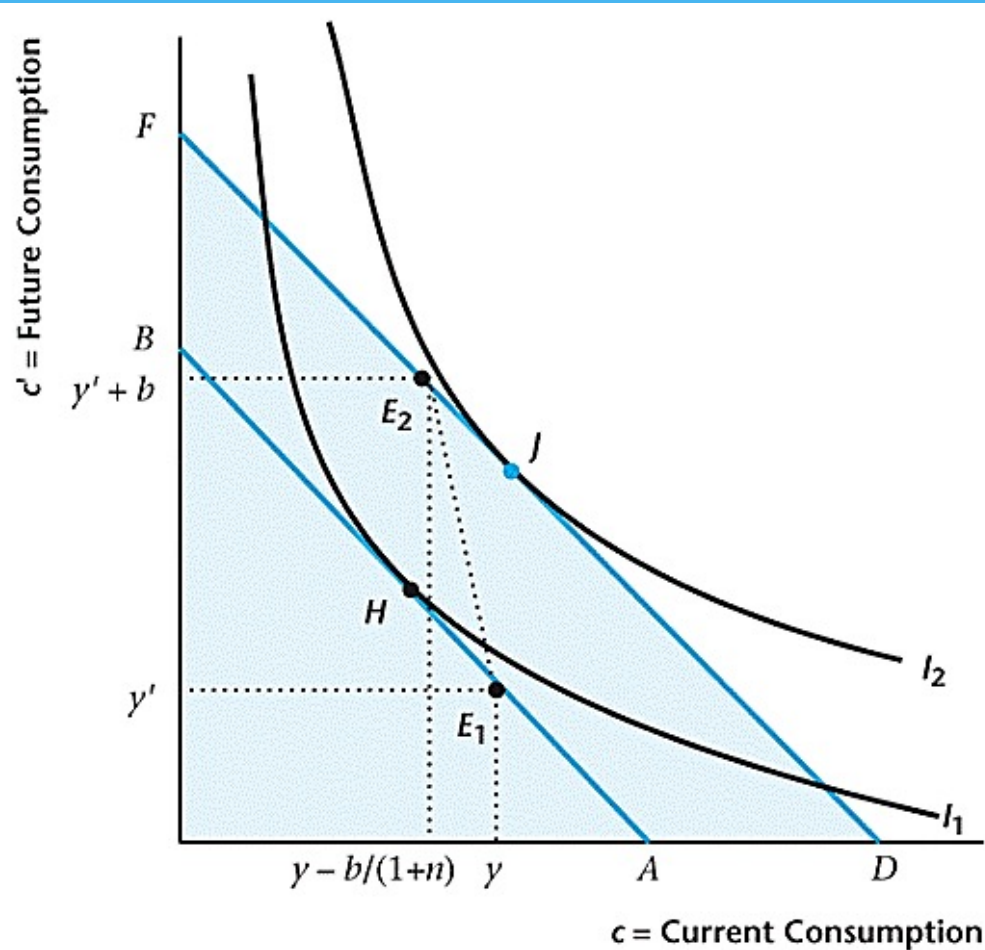


Figure 10.9 Pay-As-You-Go Social Security for Consumers Born in Period T and Later



Pay-as-you-go Social Security

- Pay-as-you-go is beneficial only if the population growth rate exceeds the real interest rate.
- The interpretation is that the population growth rate is the implied rate of return for an individual from the social security system, so social security is only worthwhile if the return exceeds what could be obtained in private credit markets.

Fully-Funded Social Security

- * Essentially a mandated savings program where assets are acquired by the young, with these assets sold in retirement.

Figure 10.10 Fully Funded Social Security When Mandated Retirement Saving Is Binding

